

Lattice balls refute the exponent restriction in Conjecture 4.2

A counterexample to arXiv:1412.6400 as printed

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Status. Candidate full counterexample, subject to human verification. In dimension 3, the polynomial

$$P(x) = 1 + x_1^2 + x_2^2 + x_3^2$$

has finite lattice sublevel sets with cardinality comparable to $t^{3/2}$. Conjecture 4.2 permits only powers $t^{\alpha/\beta}$ with $0 < \alpha \leq \beta$, hence exponent at most 1.

1 Source and exact conjecture

The source is P. L. Combettes and D. Dũng, *Kolmogorov n -Widths of Function Classes Induced by a Non-Degenerate Differential Operator: A Convex Duality Approach* [1], arXiv:1412.6400, published in *Set-Valued and Variational Analysis* 24 (2016), 83–99.

For a real polynomial P on $\mathbb{R}^d$, the source defines

$$K(t) = \{k \in \mathbb{Z}^d : |P(k)| \leq t\}.$$

Conjecture 4.2, in Section 4.6 on source PDF page 19, states:

If $K(t)$ is finite for every $t \geq 0$, then there exist integers α, β, ν with

$$0 < \alpha \leq \beta, \quad 0 \leq \nu < d,$$

such that, for all sufficiently large t ,

$$\#K(t) \asymp t^{\alpha/\beta} (\log t)^\nu. \tag{1}$$

Here $A(t) \asymp B(t)$ means that their ratio stays between two positive constants for all sufficiently large t . Figure 1 shows the exact printed statement.

The restriction $\alpha \leq \beta$ forces $\alpha/\beta \leq 1$. This is violated already by ordinary lattice balls.

2 Proof intuition

Use a positive coercive polynomial whose sublevel sets are dilates of a fixed full-dimensional body. In d dimensions, a degree- m polynomial of the form

$$1 + x_1^m + \cdots + x_d^m \quad (m \text{ even})$$

has sublevel sets of radius $t^{1/m}$. A full-dimensional lattice ball of that radius contains and is contained in coordinate cubes of comparable side length, so it has about $(t^{1/m})^d = t^{d/m}$ lattice points. Taking $d > m$ makes the power larger than one, while every sublevel set remains finite. Thus the example satisfies the sole hypothesis of the conjecture and violates its permitted conclusion.

$$d_n(U^{\lfloor r_{31} \rfloor}, L_2(\mathbb{T}^z)) \asymp n^{-z} (\log n)^z. \quad (4.31)$$

4.6 A conjecture

Suppose that $U_2^{[P]}$ is compact in $L_2(\mathbb{T}^d)$. In view of Lemma 2.9 this is equivalent to the conditions:

- (i) For every $t \in \mathbb{R}_+$, $K(t)$ is finite.
- (ii) $\tau > 0$.

As mentioned in (3.30), for every $n \in \mathbb{N}$ sufficiently large, if $t(n) \in \mathbb{R}_{++}$ is the maximal number such that $\text{card } K(t(n)) \leq n$, then

$$d_n(U_2^{[P]}, L_2(\mathbb{T}^d)) \asymp \frac{1}{t(n)}. \quad (4.32)$$

This means that the problem of computing the asymptotic order of $d_n(U_2^{[P]}, L_2(\mathbb{T}^d))$ is equivalent to the problem of computing that of $\text{card } K(t)$ when $t \rightarrow +\infty$. Let us formulate it as the following conjecture.

Conjecture 4.2 Suppose that, for every $t \in \mathbb{R}_+$, $K(t)$ is finite (the condition $\tau > 0$ is not essential). Then there exist integers α , β , and ν such that $0 < \alpha \leq \beta$, $0 \leq \nu < d$, and, for t large enough,

$$\text{card } K(t) \asymp t^{\alpha/\beta} (\log t)^\nu. \quad (4.33)$$

Figure 1: Conjecture 4.2 on page 19 of the source paper [1].

3 Counterexample

Theorem 1 (An infinite counterexample family). *Let $m \geq 2$ be even and let $d > m$. Define*

$$P_{d,m}(x_1, \dots, x_d) = 1 + \sum_{j=1}^d x_j^m.$$

Then $K_{d,m}(t) = \{k \in \mathbb{Z}^d : |P_{d,m}(k)| \leq t\}$ is finite for every $t \geq 0$, but

$$\#K_{d,m}(t) \asymp t^{d/m}. \quad (2)$$

Consequently $P_{d,m}$ refutes Conjecture 4.2 as printed.

Proof. Because m is even,

$$P_{d,m}(k) = 1 + \sum_{j=1}^d |k_j|^m \geq 1.$$

Thus $K_{d,m}(t)$ is empty for $t < 1$, while for $t \geq 1$ it is the lattice ℓ_m -ball

$$K_{d,m}(t) = \left\{ k \in \mathbb{Z}^d : \sum_{j=1}^d |k_j|^m \leq t - 1 \right\}. \quad (3)$$

In particular every such set is finite.

Put $R = (t - 1)^{1/m}$. The elementary inclusions

$$\left[-\frac{R}{d^{1/m}}, \frac{R}{d^{1/m}} \right]^d \subseteq \left\{ x \in \mathbb{R}^d : \sum_{j=1}^d |x_j|^m \leq R^m \right\} \subseteq [-R, R]^d \quad (4)$$

give, after counting integer points,

$$\left(2 \left\lfloor \frac{R}{d^{1/m}} \right\rfloor + 1 \right)^d \leq \#K_{d,m}(t) \leq (2\lfloor R \rfloor + 1)^d.$$

For sufficiently large t , both outer expressions are constant multiples of R^d . Hence

$$\#K_{d,m}(t) \asymp R^d = (t - 1)^{d/m} \asymp t^{d/m},$$

proving (2).

Suppose the conclusion of Conjecture 4.2 held. Set $r = \alpha/\beta$. Since $0 < \alpha \leq \beta$, we have $0 < r \leq 1$; also $0 \leq \nu < d$. But $d/m > 1$, and therefore

$$\frac{\#K_{d,m}(t)}{t^r (\log t)^\nu} \asymp \frac{t^{d/m-r}}{(\log t)^\nu} \rightarrow \infty.$$

This contradicts the upper comparison required by (1). □

Corollary 2 (Smallest simple instance). *For*

$$d = 3, \quad P(x_1, x_2, x_3) = 1 + x_1^2 + x_2^2 + x_3^2,$$

every $K(t)$ is finite and $\#K(t) \asymp t^{3/2}$. Hence this single quadratic polynomial refutes Conjecture 4.2.

4 Verification, interpretation, and novelty

No external theorem or numerical computation is used. The hypothesis is verified directly by coercivity, and the two cube inclusions (4) prove the exact growth order. A reviewer only needs to check that Conjecture 4.2 indeed includes the printed restriction $0 < \alpha \leq \beta$; both the arXiv PDF and the authors' currently available published-version PDF contain that restriction.

The counterexample is deliberately scoped to the statement as printed. If the intended conjecture merely asked for an arbitrary positive rational power, with no requirement $\alpha \leq \beta$, this family would satisfy the corrected conclusion by taking $\alpha = d$, $\beta = m$, and $\nu = 0$. Therefore the packet does not refute that natural corrected conjecture.

A bounded search through 9 August 2026 used arXiv:1412.6400, the exact paper title, the authors' names, "Conjecture 4.2", "card $K(t)$ ", and the displayed α/β formula. It located the arXiv record, the 2016 publication metadata, and the authors' published-version PDF, which still prints the same inequality. No erratum, later correction, or paper stating this counterexample was found. Novelty is plausible, but the most likely interpretation is a persistent typographical restriction rather than a failure of the broader lattice-counting program.

Human-review recommendation. Verify whether the authors intended to omit $\alpha \leq \beta$ or to reverse it. As a mathematical statement, Conjecture 4.2 in both located versions is false by Theorem 1.

References

- [1] P. L. Combettes and D. Dũng, *Kolmogorov n -Widths of Function Classes Induced by a Non-Degenerate Differential Operator: A Convex Duality Approach*, Set-Valued and Variational Analysis **24** (2016), 83–99; arXiv:1412.6400.